Ranga Baminiwatte

PhD Candidate in Biomedical Data Science & Informatics

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LinkedIn | ResearchGate | Google Scholar

Applied machine learning researcher focusing on AI methods that support biomedical discovery and clinical interpretation. Experienced in developing data-driven solutions for disease understanding and building end-to-end deep learning pipelines across graph learning and computer vision. Interested in developing multimodal phenotyping methods for clinical workflows.

TECHNICAL SKILLS

Programming languages | Python· R · C++ · JavaScript · SQL
Frameworks | Pytorch · TensorFlow ·Scikit-learn · PyTorch Geometric · Deep Graph Library (DGL) · Huggin Face Transformers
Tools | Linux · SLURM · Flask · PostgreSQL · Git
Domains | Graph ML·Statistical Modeling·Contrastive Learning·Phenotyping· Applied genomics·Computer Vision·NLP

RESEARCH EXPERIENCE

Graduate Research Assistant | Clemson University | Clemson, South Carolina, United States

Aug 2023–

- Design AI and statistical methods for disease understanding, multimodal phenotyping, and biomedical knowledge representation using medical images, clinical features, and knowledge graphs.
- Build end-to-end deep learning pipelines (data preprocessing, GPU-optimized training, hyperparameter search, evaluation, and reporting) for scalable, production-oriented research prototypes.
  Selected Projects:

  1. PhenoGnet - Developed a graph contrastive learning framework over multi-relational biomedical knowledge graphs to model disease similarity for patient retrieval and phenotype-based cohort discovery. Repo)

  2. (Ongoing) Developed a CNN+GCN framework to build a quantifiable ontology of facial features from clinical images and anatomical priors, supporting rare disease diagnosis and next-generation phenotyping.

Research Assistant | Higher School of Economics | Moscow, Russia

Jan 2023–Aug 2023

- Developed machine learning models for scientific and applied research questions in linguistics and physics.
- Conducted data engineering and evaluation on complex, noisy multi-sensor datasets (eye tracking, demographics, and text), designing ML models robust to real-world noise.
  Selected Projects:
  1. A Machine Learning framework for predicting the First-Language (L1) and the Second-Language proficiency using eye movements during reading and demographics information of Non-native English learners. Report | Repo

  2. Simulation of Time Projection Chamber data of a particle detector using Deep Generative models, including Generative Adversarial Networks (GANs), Variational Autoencoders(VAEs) and Normalizing Flows. Report

Student Intern | Raiffeisen Bank Russia | Moscow, Russia

Nov 2020–May 2021

- Researched blockchain-based compliance validation systems integrating smart contracts and SQL-backed databases for secure corporate settlements.
  Project: Corporate Settlement Mechanism Using Blockchain Smart Contracts – Developed Solidity contracts with a C++ R-Chain adapter and PostgreSQL integration for automated deal execution. Report

EDUCATION

Clemson University & Medical University of South Carolina | SC, USA
Aug 2023 - Jul 2027
Doctor of Philosophy - Biomedical Data Science and Informatics
Aug 2023 - Dec 2025
Master's - Biomedical Data Science and Informatics (Master's en route to PhD)

Higher School of Economics | Moscow, Russia
Sep 2018 - Jul 2023
Bachelor of Science - Applied Mathematics and Computer Science (Track: Data Science and Business Analytics)

SELECTED PUBLICATIONS/ PRE-PRINTS

1. **Baminiwatte, R.**, Rana, K.J., Masino, A.J.: PhenoGnet: A Graph-Based Contrastive Learning Framework for Disease Similarity Prediction. *arXiv preprint arXiv:2509.14037* (2025) link

2. Masino, A.J., Baminiwatte, R.: Automated Shared Phenotype Discovery in Undiagnosed Cohorts for Rare Disease Research. In: 2024 International Conference on Machine Learning and Applications (ICMLA). pp. 1025–1030. IEEE (2024) link

3. **Baminiwatte, R.**, Torsu, B., Scherbakov, D., Mollalo, A., Obeid, J.S., Alekseyenko, A. V, Lenert, L.A.: Machine learning in healthcare citizen science: A scoping review. Int J Med Inform. 105766 (2024) link

4. Shalileh, S., Kairov, M., Baminiwatte, R., Parshina, O., Dragoy, O.: Predicting First-Language and Second-Language Proficiency Using Eye Fixation Data and Demographic Information: Assumptions, Data Representations and Methods. IEEE Access. (2024) link

CONFERENCES, TALKS & PRESENTATIONS

1. Baminiwatte, R. et al. "AI methods for next-generation phenotyping in Biomedical Applications" **Poster** to be presented at Annual Symposium of American Medical Informatics Association(AMIA), Atlanta, Georgia, Nov 2025.
2. Baminiwatte, R. et al. "AI methods for next-generation phenotyping in rare genetic disorders" **Poster** presented at the Annual Meeting of the American Society of Human Genetics(ASHG), Boston, Massachusetts, Oct 2025.
3. Baminiwatte, R. et al. "AI methods leveraging formal biological knowledge for next-generation phenotyping in rare genetic disorders", **Poster** presented at the Clemson High Performance Computing (HPC) day, Clemson, SC, 17 Apr 2025.
4. Baminiwatte, R. et al. "AI methods leveraging formal biological knowledge for next-generation phenotyping in rare genetic disorders" **Talk** delivered at the Clemson Center for Human Genetics, Greenwood, SC, 15 Sep 2025.
5. Baminiwatte, R. et al. "PhenoGnet: A Graph-Based Contrastive Learning Framework for Disease Similarity Prediction" **Poster** presented at the Clemson Artificial Intelligence Symposium, Clemson, SC, 17 Apr 2025.

SCHOLARSHIPS AND OTHER AWARDS

Sep 2025 'Excellence in Communications Award' – Clemson University High Performance Computing Poster Competition
Aug 2023 Graduate Research Assistantship - Clemson University
Jan 2023 Open Doors: Russian Scholarship project – Recipient
Aug 2017 Russian Federation State Scholarship for International Students – Recipient